

31338 Network Servers

Lab 1c: Windows Server 2019

Aims

1. Setup the initial Windows Server 2019 configuration.

Task 1: Startup

The supplied VMware image of Windows Server 2019 has already been installed, but it is an evaluation version without a license key. It will work for a finite time in evaluation mode.

Use VMware Player to start your WindowsServer server (see Lab01a for instructions on how to do this).

One of the first things you should notice is that the startup sequence is different from Linux. You can't press ESC to display the bootup logging, so just be patient.

Windows Server 2019 will first boot up the GUI with a prompt to enter the administrator password. You need to press Ctrl-Alt-Delete to bring up the password prompt but this can be problematic because your host system will pick up the key sequence rather than the guest VM. One option is to press Ctrl-Alt-Insert instead. If your keyboard doesn't have a dedicated 'Insert' key, the other option is to use the VMware menu to choose "VM" → "Send Ctrl+Alt+Delete".

Enter the password below for the Administrator user:

```
student123!
```

You should now get a desktop, and the "Server Manager > Dashboard" panel. Click on "Local Server" – this is your Windows Server 2019 machine.

Check/change the following information

1. Make sure the time zone is set to (UTC+10:00) Canberra, Melbourne, Sydney – change if it's not.
2. Check the networks – Ethernet0 and Ethernet1 should be IPv4 address assigned by DHCP, IPv6 enabled. You can click on the "IPv4 address assigned by DHCP" which will open the more detailed Network Connections configuration page.
3. Change the computer name and description to something more reasonable, for example, "chrisserver", but for the moment, leave the workgroup as "WORKGROUP". You will likely be asked to reboot the server. You can do that now, or in a little while when you are finished exploring.
4. Note the section on updates to see when updates were last installed/checked, and how Windows Update handles updates. Note that a server is different to a workstation – you don't want the server rebooting itself unexpectedly due to updates being installed.

Make a note in your journal the details you changed, and any other configuration you find noteworthy.

Scroll down to view events (logs), services, and other information about your local server.

If you didn't reboot your server already (due to changing its name), do that now before proceeding.

Task 2: Server Management

As well as a quick overview of the server's configuration and status, the Server Manager is the main console for managing many aspects of your server. It allows you to:

1. View (and update) the server summary – including information about the computer, security information
2. View (and update) the server roles
3. View (and update) the server features
4. Summarises common resources and support links and has the 'Best practices analyzer'

There is a “Tools” menu up the top of the Server Manager that allows you to open and run additional server management tools:

5. View Diagnostic information such as the event viewer and device manager
6. View (and update) the configuration such as Task Scheduler, Firewall, Services, Computer Management
7. View the backup and disk management tools

Activities: We will explore the server manager by the following activities

1. Explore the device manager (you have to find it first ☺). What hardware is installed on this machine? Is it similar to the Linux image?
Write this down in your journal.
2. What is your network address (IP address) that connects to the internet? Can you check it against the Linux server? What is the subnet and gateway?
3. Add a feature:
See if you can add the “Telnet Client” and the “Simple TCP/IP services” features. This starts some common TCP/IP services such as the echo and daytime services. **Don’t** change the Server Roles at this time.

Check that this is running by viewing the services panel (your job to find this one!) and scrolling down the services panel to find the Simple TCP/IP services item

If it isn’t started, see if you can work out how to start it (hint: green arrow, or the “More Actions” panel on the left, or right-click.

4. You can test this by starting a Command Prompt from the Start menu (under “Windows System” folder) and typing:
telnet localhost 13
telnet localhost 17

(Note: even if you know about PowerShell, the Command Prompt is better for this exercise so you can see the output of telnet).

5. From your host workstation, try using the above commands to your server’s IP address. Can you do this? Why do think this is so?

Task 3: Command line

Virtually all of the commands to setup Windows Server 2019 can be done from the command line. Here are some samples:

net	contains many functions for managing the local server
netdom	contains many functions for managing the domain
netsh	contains many functions for managing the local network

Use

net help *command* to see the syntax of the commands

Activities:

1. Let’s see what services are running.
net start
Observe the output. Is Simple TCP/IP Services running? You can stop them using the net stop command
net stop “Simple TCP/IP Service”

Test that this is not working anymore (hint: telnet ?)
Restart the service (hint: net start)

2. Let’s allow the “quote of the day” though the firewall (the whole command below should be on one line):
**netsh advfirewall firewall add rule name="TCP Port 17" dir=in
action=allow protocol=TCP localport=17**

Try the earlier exercise by telnet-ing from your host workstation to port 17.
Hopefully it worked this time!

Use your journal and the internet to view the syntax of these commands.

Task 4: Documentation

Windows commands generally have excellent contextual documentation.

For example, you can learn more about how to use Server Manager using the “Help” menu.

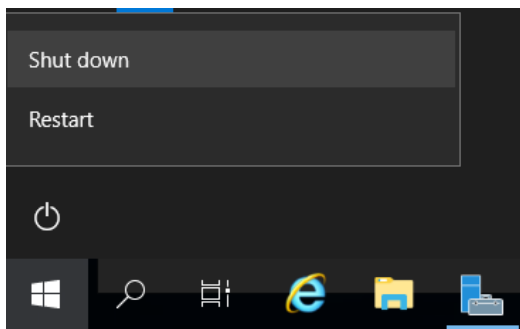
The Server Manager also provides links to Microsoft documentation, plus Google can be your friend.

Notice that Internet Explorer has an Enhanced Security Configuration. You may get prompts to Add the site – the decision is yours, but generally you can cancel them (this is usually blocking ActiveX, flash or popup windows). If it seems to block too much, you can adjust the settings for the Enhanced Security Configuration (opening a fresh Internet Explorer window should give you links showing you how).

Make notes in your journal about what you find, and any changes you make to the Internet Explorer Enhanced Security Configuration.

Task 4: Shutdown

Finally shutdown your VM via the Start Menu.



Note that Windows Server requires you to enter a reason for the shutdown.

This is generally good practice to enter something reasonable, so an audit or the next admin can see why you shut the server down.

Choose a topic (eg: Other(Planned)) and a reason and shutdown the system.

HINT: If you need to temporarily suspend your work, you can use VMware’s suspend feature (this is under the VMware menu: VM → Power → Suspend).

This saves a snapshot of your memory in a file called *.vmem and runtime details in the *.vmss file.